

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : **C4**

Team Members :

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1. Problem Title : ***Clinic Appointment Scheduler System***

2. Problem Description :

A Clinic Appointment Scheduler System is designed to manage and organize patient appointments in a clinic efficiently. Patients can book appointments with doctors based on available time slots. Doctors can manage their schedules and view upcoming appointments. Receptionists can assist in booking, cancelling , or rescheduling appointments.

The system reduces manual errors, improves scheduling efficiency, and ensures smooth communication between patients, doctors, and clinic staff.

3. Identified Entities (Classes)

Class Name	Description
Patient	Represents a person who books appointments with doctors
Doctor	Represents doctors who provide consultation in the clinic
Appointment	Stores details of a scheduled appointment
Receptionist	Handles appointment booking and management
Schedule	Manages available time slots for doctors
Clinic	Represents the overall clinic system managing doctors and appointments

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Patient	patientID, name, phoneNumber, age	bookAppointment(), cancelAppointment(), viewAppointment(), updateProfile()
Doctor	doctorID, name, specialization, availableSlots	viewAppointment(), updateAvailability(), acceptAppointment(), cancelAppointment()
Appointment	appointmentID, patientID, doctorID, appointmentDate, appointmentTime	scheduleAppointment(), cancelAppointment(), rescheduleAppointment(), confirmAppointment()
Receptionist	receptionistID, name, contactNumber	bookAppointment(), cancelAppointment(), generateDailySchedule()
Schedule	scheduleID, doctorID, availableSlots	checkAvailability(), addTimeSlot(), removeTimeSlot()
Clinic	clinicName, location, doctorList, appointmentList	addDoctor(), registerPatient(), manageAppointments(), generateReports()